General Instructions

WARNING

Detailed information on standard workshop and safety procedures, and general servicing operations is not included in this manual, which has been prepared to assist qualified service personnel. ARGO assumes no responsibility or liability for PERSONAL INJURY or VEHICLE DAMAGE which results from any servicing procedure performed, including those instructions outlined in this manual.

Automatic Torque Converter System

The torque converter system automatically regulates the torque delivered by the engine to the transmission. The System consists of a driver clutch located on the engine output shaft, a driven clutch located on the input shaft of the transmission, and a drive belt.

On acceleration, the driver clutch moveable face travels towards the fixed face as the weights in the clutch are pushed outwards due to centrifugal force. This in turn compresses the internal spring of the driver clutch. As the driver clutch faces narrow, they grip the drive belt moving it towards the top of the driver clutch. (Top No Load Speed). At the same time, the transmission's driven clutch plates spread apart as the belt rides down between the faces. The tension in this clutch is also regulated by a spring. As the vehicle comes under load, the spring loaded cam on the driven clutch forces the clutch plates together, causing the CVT to "Back Shift".

NOTE

At idle, the belt should rest flush or a little beyond the edge of the driven clutch on the transmission. At top no load speed the belt should run near or to the top of the driver clutch.

CAUTION

Worn or dirty clutch components, or a worn drive belt will make the ARGO unreliable and unsafe to operate.

Maintenance Information

The maintenance information provided in this manual refers to an ARGO vehicle operating under normal conditions and use. Service more often when the vehicle is used under heavy duty applications.

Servicing of the Automatic Torque Converter system is necessary when:

A drop in vehicle performance occurs:

- the vehicle does not speed up or slow down smoothly
- the vehicle hesitates or sticks at one speed
- the vehicle vibrates severely during normal operations
- the vehicle does not move when placed in gear
- the vehicle creeps at idle when placed in gear
- the vehicle is hard to shift into gear or other ranges

Drive Belt Maintenance Schedule-Inspect every 25 hours

WARNING

Do not attempt to service any part of the torque converter system while the engine is running. Shut the engine off and disconnect the battery before servicing.

NOTE

Drive belt tension and clutch alignment are fixed. No further adjustment is required.

WARNING

Do not attempt to adjust, repair or replace the drive belt, clutches or any moving part while the engine is running. Doing so will cause injury. Before servicing the vehicle, disconnect the battery to prevent accidentally starting the engine.

Keep the engine compartment hood, clutch guard (if applicable) and firewall securely in place when the engine is running. Severe injury can result if the drive belt, clutch components or other moving parts come loose.

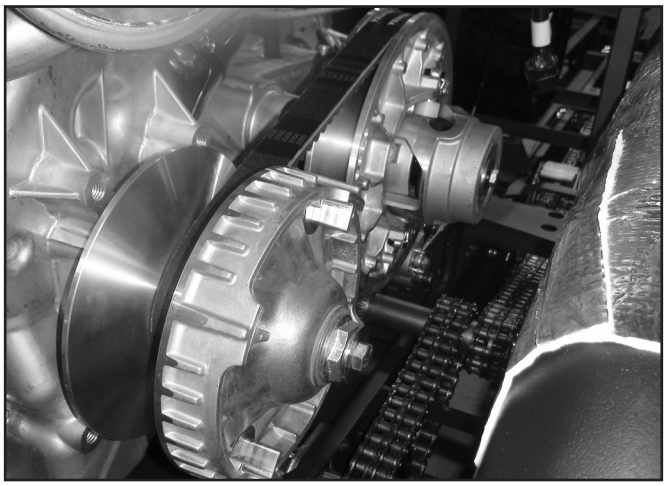
If engine compartment inspection is necessary while the engine is running, use EXTREME CAUTION! Keep engine RPM low. Avoid standing directly in line with moving components. Use a mirror to view the components.

Drive Belt Inspection

The drive belt transmits power from the driver clutch (on the engine) to the driven clutch (on the transmission). These components are located on the left side of the engine compartment.



Clutch System (CVTech)



Instance Torque Drive System (ITDS)

Check the drive belt after every 25 hours of operation, or whenever there is a noticeable reduction in clutch performance. Replace the belt when:

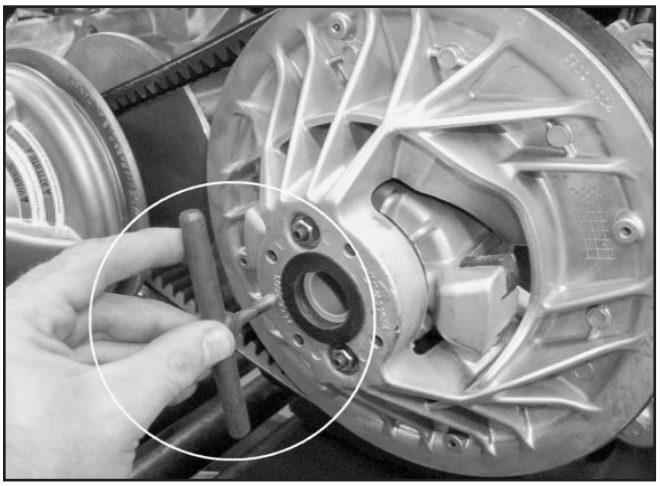
- the top width of the belt has worn to 1-1/16" (27mm)
- cracks, fraying or shredding is apparent
- it becomes contaminated with oil or some other fluid

Check the condition of the belt in the table below and perform the necessary servicing procedure.

Belt Condition	Problem	Servicing Procedure
Glazed Sides	Lubricant Contamination Clutch Malfunction	Replacing the Drive Belt Driver Clutch Servicing Driven Clutch Servicing
Torn Underchords	Wrong Drive Belt Old Drive Belt	Replacing the Drive Belt
Worn Prematurely	Driver or Driven Clutch Problem	Driver Clutch Servicing Driven Clutch Servicing Replacing the Drive Belt

Removing The Drive Belt (CVTech/Invance)

1. Perform the servicing procedure, **Removing The Firewall.**
2. Locate the 6mm x 1.0 threaded hole in the clutch face. This hole is provided to assist in spreading the driven clutch sheaves by threading a 6mm x 1.0 thread bolt in through the face. This bolt should be a least 2" in length with full thread.
3. Thread the bolt into the clutch face to spread the sheaves.
4. Slip the belt off between driven and driver clutch.



Installing The Drive Belt (Invance/CVTech)

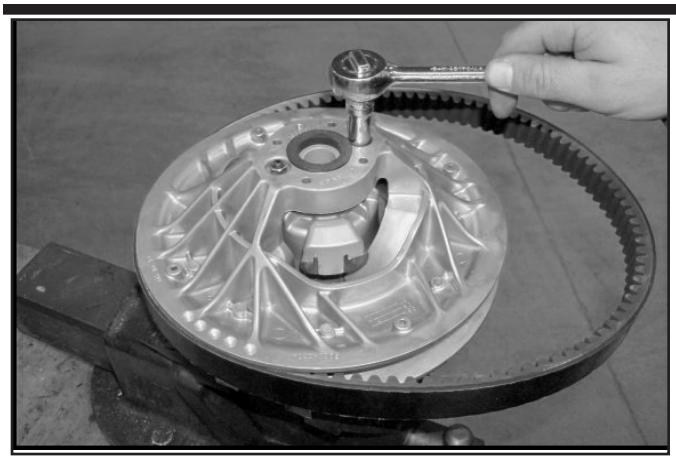
CAUTION

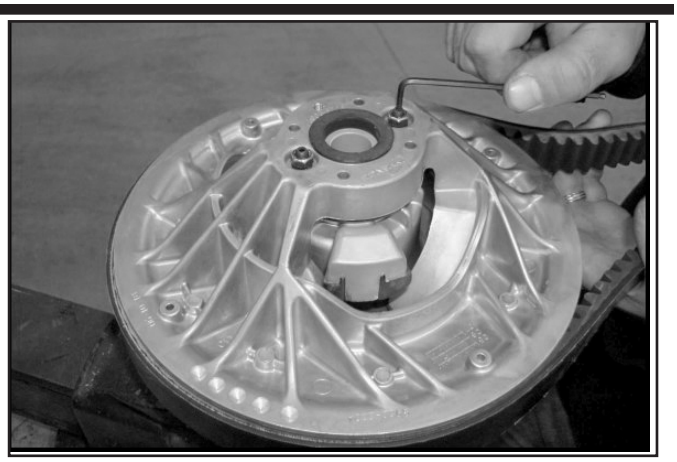
If this procedure is not carried out as described, the edge of the fixed face may cut or damage the drive belt.

1. Position the belt around the driver clutch first.
2. Ease the belt over the edge of the fixed face on the driven clutch and at the same time, turn the inside, movable face clockwise.
3. Remove the 6mm x 1.0 thread bolt from the clutch face to allow the driven clutch sheaves to return to the closed position.

Invance Driven Clutch Drive Belt Adjustment

1. Perform the servicing procedure, **Removing The Drive Belt (Invance Driven Clutch)**
2. Perform the servicing procedure, **Removing The Driven Clutch (Invance).**
3. Remove the driven clutch from the vehicle and move to a clean work bench to loosen both jam nuts located on the fixed face of the clutch.
3. Using an allen wrench, turn the adjustment set screw either in (to lower the belt) or out (to raise the belt) between the clutch sheaves.

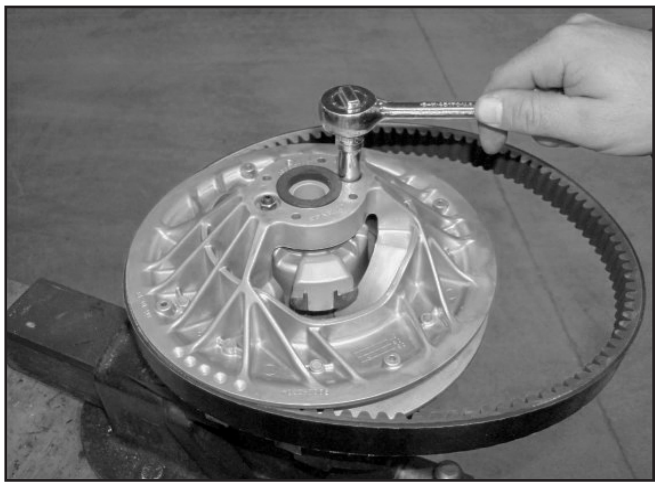




⚠ IMPORTANT

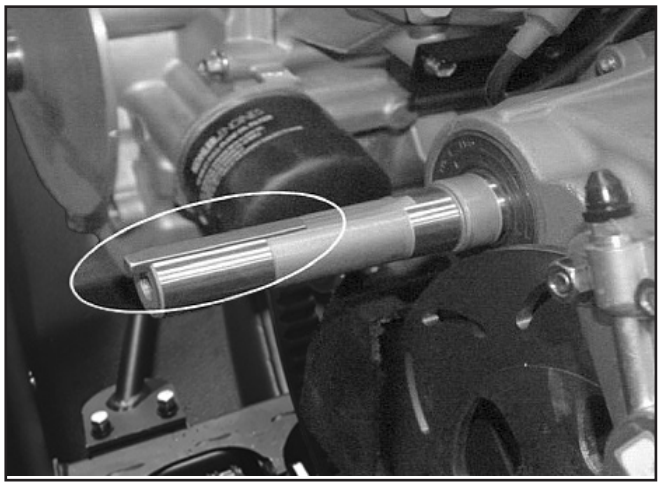
Loosen set screws uniformly 1/2 turn at a time. It is crucial that the clutch faces remain true and parallel to each other around the entire circumference of the clutch.

4. After adjustment, check belt level by placing a drive belt between the sheaves. Belt position should be anywhere from flush with the top of the sheaves, to 1/10" above. Re-tighten jam nuts and torque to 60-75 in. lbs. (7.5 +/- 1 Nm)



Removing The Driven Clutch (Invance)

1. Perform the servicing procedure, **Removing The Firewall.**
2. Perform the servicing procedure, **Removing The Drive Belt (CVTech/ Invance)**
3. Locate the HHCS 3/8-16x3.5 Gr5 Zn, Lock Washer and Flat Washer securing the clutch to the input shaft and remove with 9/16" socket wrench.
4. The Invance Driven Clutch is installed to a straight keyed shaft. Slide the Invance driven clutch off the input shaft.



Installing The Driven Clutch (Invance)

1. Install the key to the keyway of the input shaft and apply anti-seize compound along the length of the shaft.
2. Apply Blue 242 LOCTITE to the HHCS 3/8-16x3.5 Gr5 Zn, and secure the clutch along with Lock Washer and Flat Washer. **Torque fastener to 15 FT. LBS. 20.3Nm**
3. Perform the servicing procedure, **Installing The Drive Belt (Invance/CVTech).**

Clutch Maintenance

Disassembly and repair of the driver and driven clutch requires special tools. The following indicates that clutch service might be required:

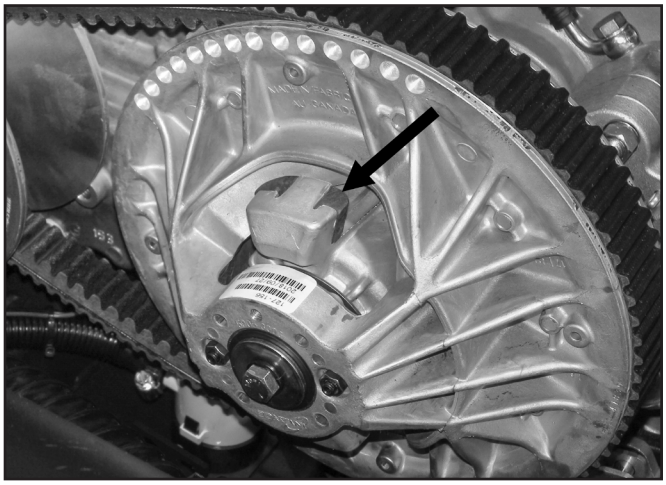
- a drop in vehicle performance
- the clutch does not shift smoothly
- the clutch sticks during vehicle operation
- the drive belt wears rapidly
- the vehicle vibrates severely during operation
- the vehicle does not accelerate when the engine speed is increased with the transmission in gear
- transmission will not shift smoothly into gear at engine idle.

Driven Clutch Inspection

Inspect the nylon sliders every 50 hours. The nylon sliders are mounted in the driven clutch moveable pulley. When the clutch shifts, the cam moves on the nylon sliders. Replace the nylon sliders before there is aluminum to aluminum contact between the cam and the movable pulley. Driven clutch disassembly is required to replace the nylon sliders properly.

Driven Clutch Lubrication

Lubrication of Driven Clutches is not necessary



(CVTech) Invance Driven Clutch Sliders



ITDS Driven Clutch Sliders

Driver Clutch Inspection

If the vehicle is not shifting speeds smoothly, if the vehicle hesitates or sticks at one speed, or if the vehicle creeps at idle or is hard to shift into gear, remove and inspect the driver clutch. Service as necessary.

Remove and service the driver clutch if:

- the clutch plates have tracking grooves caused by drive belt action
- the clutch plates are bent or cracked
- the weight blocks are damaged or missing

Driver Clutch Removal (CVTECH)

1. Remove the engine access cover.
2. Perform the servicing procedure, **Removing the Firewall**,
3. Perform the servicing procedure, **Removing the Drive Belt.**
4. Remove the fastener securing the Driver Clutch to the PTO and slide the clutch from the shaft.

⚠ NOTE

The driver clutch must be slid from the PTO with the mounting bolt hardware within the clutch. The mounting bolt is too long to remove first.

The driver clutch may contact the lower body first before completely being free of the PTO, however the lower body will flex enough in that area to free the clutch.

5. Remove the clutch to a clean work area for further disassembly.

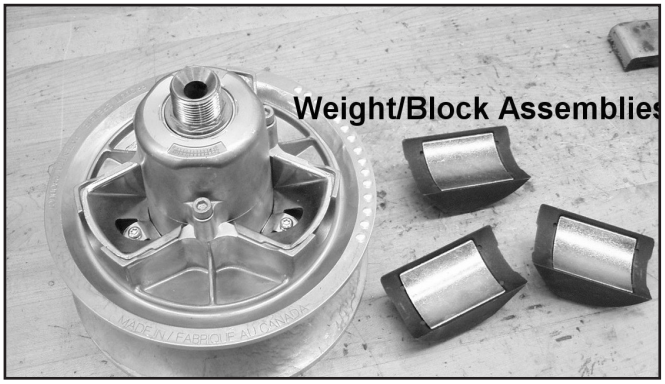
⚠ CAUTION

Be aware of the clutch washer/spacer installed to the PTO shaft behind the clutch. Ensure this washer is in place when the clutch is reinstalled or damage to engine or clutch may result when the engine is restarted.

Disassembling the Driver Clutch (CVTECH)

1. Remove the large hex nut, flat washer and cap.
2. Remove the (3) nylon blocks and weights. Weight grams will vary between models.



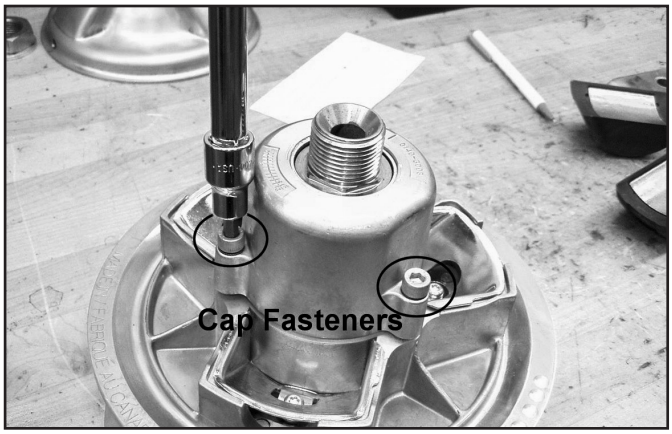


3. Remove the (3) fasteners securing spring cover to the sliding flange.

CAUTION

Cap is under spring tension. Slowly unthread each fastener uniformly and a little at a time until the cap is freed from the sliding flange.

4. Remove the cap, spring and spring guide washer.





5. Slip off the Sliding Flange.



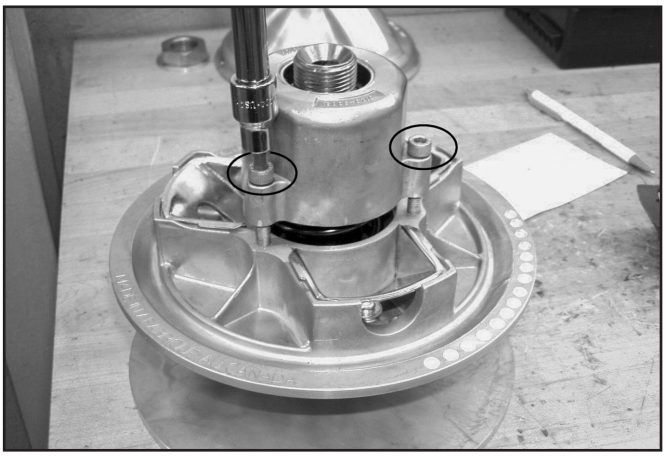
Re-assembling the Driver Clutch (CVTECH)

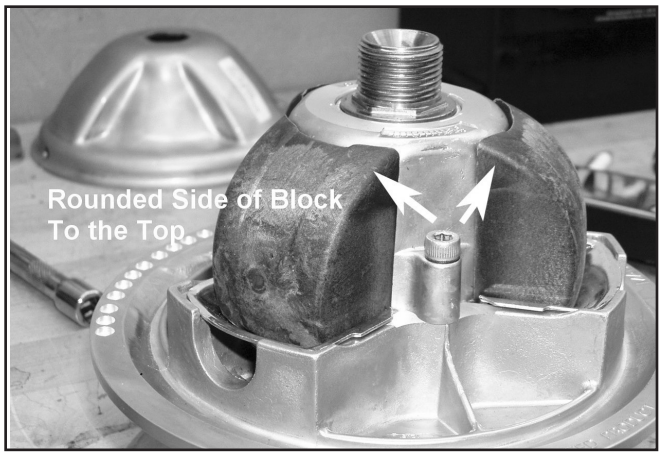
1. Slip on the Sliding Flange followed by the spring guide washer.
2. Install the spring.
3. Install the spring cap and secure with the (3) fasteners. **Torque to 85 in. lbs. (9.6Nm)**

IMPORTANT

There are (3) small lock washers installed with these (3) fasteners.

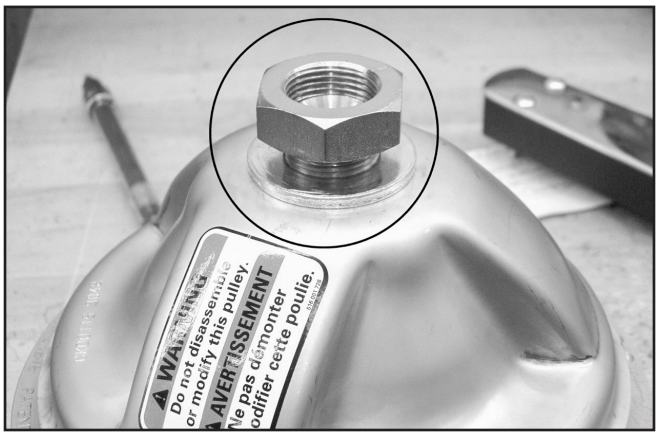
4. Re-install the nylon block and weight assemblies into the sliding flange. Orient the block with the rounded side up.
5. Re-install the cap and secure with flat washer and nut. **Torque to 100 FT. LBS. 135.6Nm**





NOTE

To properly torque, re-install the driver to the engine PTO and lock flywheel while using the torque wrench to achieve proper torque.



Install The Driver Clutch (CVTECH)

1. Ensure the spacer is installed to the PTO shaft first before installing the driver clutch.
2. Place the clutch key into the keyway of the engine PTO.
3. Apply anti-seize compound to the PTO shaft.

4. Apply 242 Blue LOCTITE to the threads of the mounting bolt and insert it along with the washer hardware into the clutch first before sliding on to the PTO shaft.
5. Align the keyway of the clutch with the key installed to the PTO shaft.
6. Slide the clutch on to the shaft and up to the shoulder of the PTO shaft.

CAUTION

The clutch should slide on by hand. Do NOT use force or tap on with a hammer.

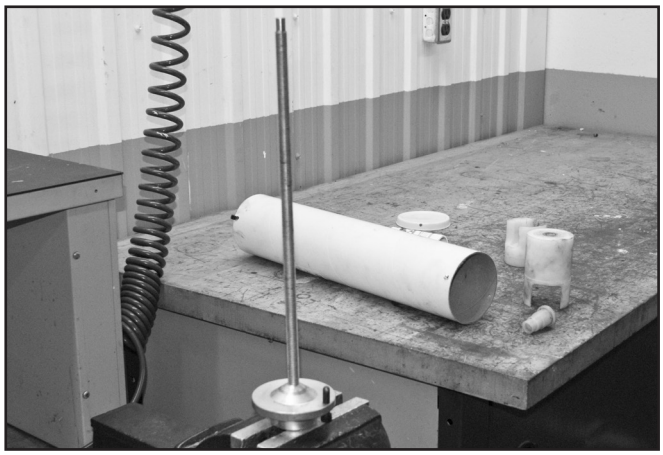
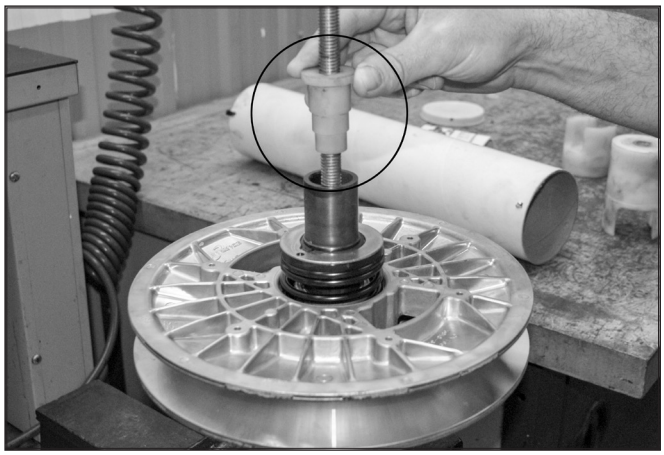
7. **Torque to 40 FT.LBS.**

Disassembling The Driven Clutch (Invance)

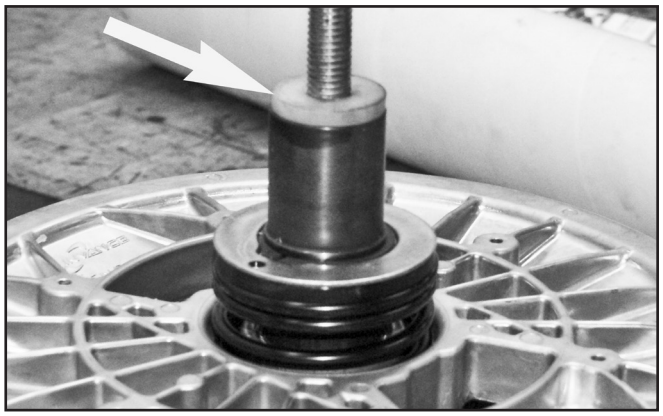
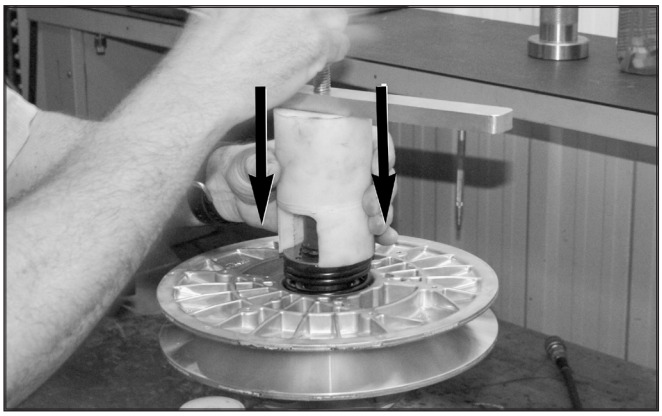
1. Remove the engine access cover.
2. Perform the servicing procedure, **Removing the Firewall**.
3. Perform the servicing procedure, **Removing The Drive Belt (Invance).**
3. Inspect the clutch and perform the servicing procedure, **Disassembling The Driven Clutch (Invance)** if:

- the clutch plates have tracking grooves caused by drive belt action
- the clutch plates are cracked or bent
- the nylon cam slider shoes are missing, melted, or worn through
- the spring is broken or has come loose from the cam or moveable face
- is not changing speed smoothly
- hesitates or sticks at one speed
- lacks turning power
- won't move when the engine is accelerated in gear and brakes are disengaged
- does not backshift under load

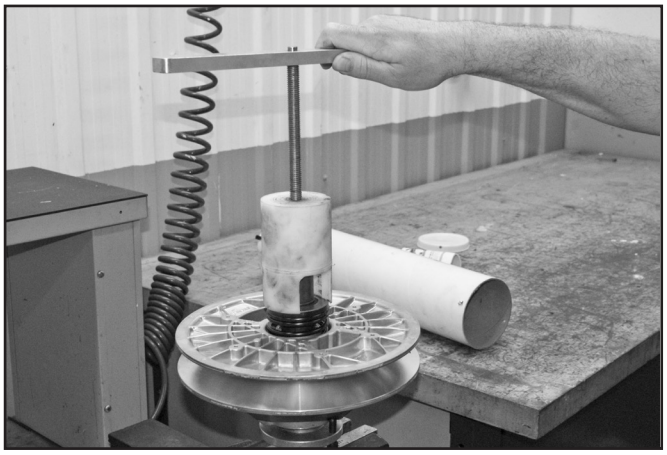
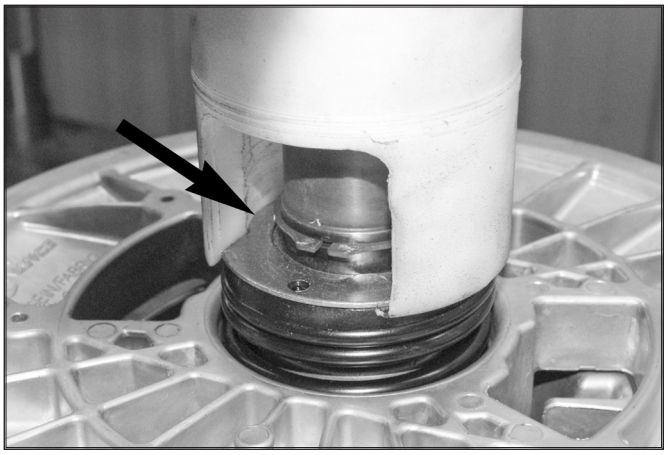
1. Place the Specialized Clutch Tool, 5055-0002, into an aluminum jawed vise.
2. Place nylon bushing into clutch shaft to centre clutch on the threaded rod.

3. Slip the nylon fork over the threaded rod and rest up against the steel washer on top of the spring.

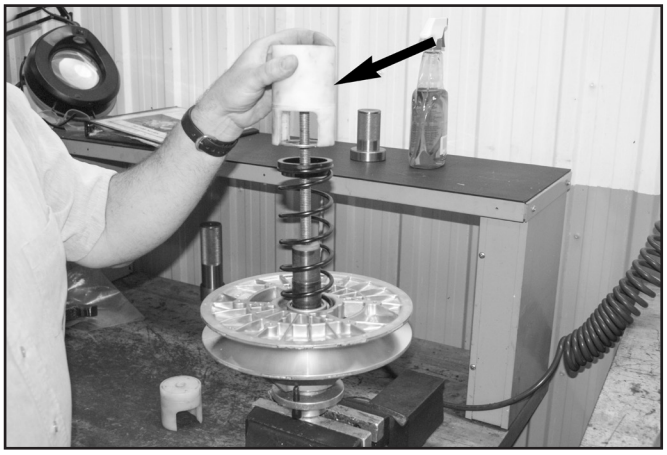
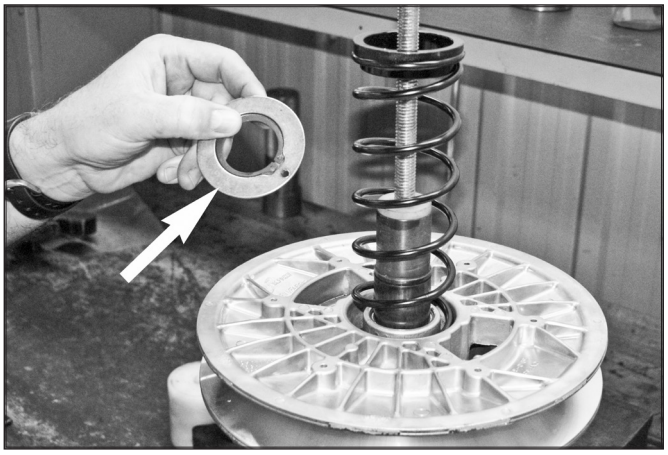



4. Thread the handle down until it makes contact with the fork and begins to compress the spring below. Thread the handle just enough to compress the spring for retaining ring removal. Back off the handle to release the tension on the spring.



5. Remove nylon fork, steel washer and spring.

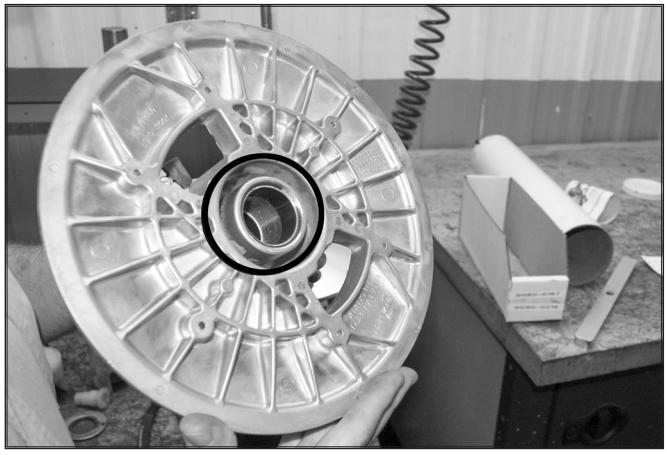
6. Be aware of the Plastic Spring Protector assembled to the top of spring.





7. Remove the moveable face.
8. Bushings are non-replaceable in moveable face.





9. Locate cam shoes for inspection. Each one is secured by 1 screw.
10. Components of disassembled Invance Clutch: Shim Washer, Spring Support Washer, Retaining Ring and Nylon Spring Protector.

